Sharing the Cost of Connecting Renewables to the Grid

A Cooperative Game Theory Approach · Nucleolus · LP Duality · Convexity

ECON 138 / ECON 238
Stanford University · PUC-Rio

1 Cooperative Games

Cooperative game with transferable cost. To make precise how a group should share the cost of jointly operating a shared system, we need a short layer of definitions drawn from the theory of *cooperative games*. A *cooperative cost game* is a pair (N, C) where $N = \{1, \ldots, n\}$ is a finite set of players and $C : 2^N \to \mathbb{R}_{\geq 0}$ is a *cost function* assigning to each coalition $s \subseteq N$ the minimum cost $C(s)$ its members would pay if they acted alone — if they built and operated their own system without any help from outside s — with the convention $C(\emptyset) = 0$. In the Application section below, $C(s)$ is the optimal value of a linear program computed one per coalition; for the moment we treat C abstractly.

Coalitions as subsets and as binary vectors. A coalition is any subset $s \subseteq N$. For computational convenience we encode the same object as a binary vector $s \in \{0,1\}^n$, with $s_i = 1$ iff player $i \in s$. We use the two representations interchangeably: $|s| = \sum_{i \in N} s_i$ counts the members, and set operations $s \cup s'$ and $s \cap s'$ correspond component-wise to $\max(s_i, s'_i)$ and $\min(s_i, s'_i)$. This is why the same symbol s appears both as a binary index inside summations like $\sum_{i \in N} s_i \cdot x_i$ and as a subset in statements like "for all $s \subsetneq N$".

Allocation and efficiency. An *allocation* is a vector $x \in \mathbb{R}^n$ where x_i is the share of the total cost charged to player i. The allocation is *efficient* if

$$\sum_{i \in N} x_i \;=\; C(N),$$

that is, the full cost of the grand coalition is distributed among its members — no more, no less. Efficiency is a budget-balance condition: the operator neither over-recovers nor under-recovers. It still leaves enormous freedom — any point on the hyperplane $\sum_i x_i = C(N)$ is efficient — and so the central question of cooperative cost games is which efficient allocation is both *fair* and *stable*, in the sense that no sub-coalition is better off walking away and paying $C(s)$ on its own. The next section makes this question precise.

2 The Core and the Nucleolus

2.1 The Core

The **core** of the game is the set of efficient allocations $x = (x_1, x_2, \ldots, x_n)$ such that no sub-coalition has an incentive to break away. Formally:

$$\mathcal{C} = \Big\{ x \in \mathbb{R}^n \;\Big|\; \textstyle\sum_{i \in N} x_i = C(N), \quad \sum_{i \in N} s_i \cdot x_i \leq C(s) \;\; \forall\, s \subsetneq N,\; s \neq \emptyset \Big\}.$$

The core is a polyhedron. It is the intersection of finitely many linear conditions — one affine equality (efficiency) and $2^n - 2$ affine inequalities (one per proper non-empty sub-coalition). Hence $\mathcal{C}$ is a convex polyhedron in $\mathbb{R}^n$; depending on the game, it may be empty, a single point, a line segment, or a higher-dimensional polytope. In Section 3.4 we prove that for the application studied here $\mathcal{C}$ is non-empty.

Two-player example. We illustrate the core with the simplest non-trivial instance: only players 1 and 2, with $C(\{1\}) = 90$, $C(\{2\}) = 100$, $C(\{1,2\}) = 120$. The total savings from cooperation are $90 + 100 - 120 = 70$. Efficiency requires $x_1 + x_2 = 120$. Writing $x_2 = 120 - x_1$, the core constraints $x_1 \leq C(\{1\})$ and $x_2 \leq C(\{2\})$ reduce to:

$$20 \;\leq\; x_1 \;\leq\; 90.$$

The core is thus a line segment in the (x_1, x_2) plane — a one-dimensional polyhedron. Figure 1 shows the segment together with the nucleolus, a special allocation point that we will compute in the next subsection.

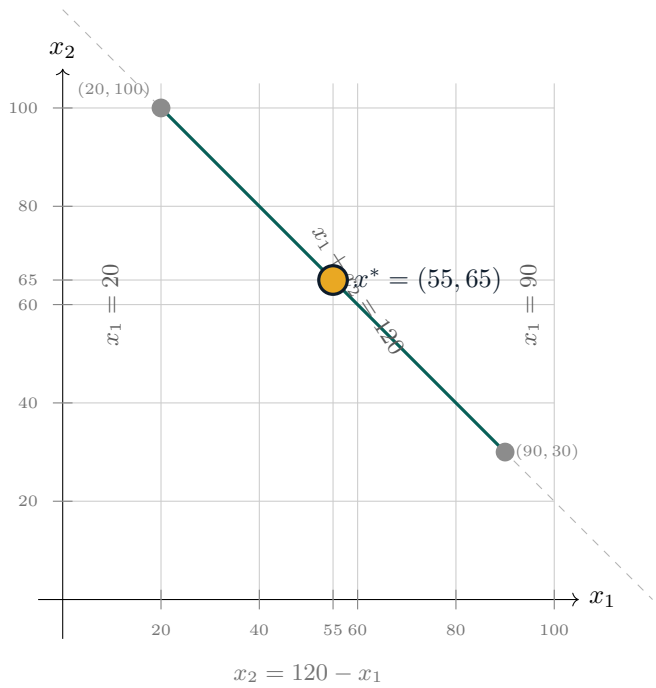

Figure 1: Two-player core (teal segment on the line $x_1 + x_2 = 120$, $x_1 \in [20, 90]$) and nucleolus $x^* = (55, 65)$ (gold point).

2.2 The Nucleolus

Why not simply pick any core point?

The core is typically a continuum of allocations — a whole polyhedron. Every point in it satisfies the rationality requirements, so no sub-coalition has a formal incentive to break away. But different core points treat players very differently. In the 2-player example above, the two endpoints of the core segment illustrate this tension: at $(90, 30)$, player 1 pays its full stand-alone cost $C(\{1\}) = 90$ and player 2 captures almost all of the cooperative savings; at $(20, 100)$, the roles are reversed. Both are "stable", yet both are highly asymmetric. A selection principle is needed to pick a *uniquely fair* point from the core.

Excess as gain

For a given allocation x and coalition s, define the **excess** as the *benefit*, or *gain*, coalition s enjoys under x relative to acting alone:

$$e(s,x) \;=\; C(s) - \sum_{i\in N} s_i \cdot x_i.$$

Here $C(s)$ is what the coalition would pay on its own, and $\sum_{i\in N} s_i \cdot x_i$ is its allocated share of $C(N)$. Therefore:

- $e(s,x) > 0$: coalition s pays *less* than its stand-alone cost — it receives a positive *benefit* from cooperation.
- $e(s,x) = 0$: the coalition breaks even — no benefit, no loss.
- $e(s,x) < 0$: the coalition pays *more* than it would alone — it has a *grievance*. (Inside the core all excesses are non-negative, so this only arises for allocations outside the core.)

Inside the core, every coalition has $e(s,x) \geq 0$, but the *distribution* of these benefits across coalitions can still be very unequal. A large $e(\{i\},x)$ means player i gets a disproportionately large share of the savings at the expense of others. *The excess is a cardinal measure of how generously a coalition is treated under allocation x.*

The maximin criterion and a first definition of the nucleolus

A natural fairness criterion is to *maximize the benefit of the worst-off coalition*: among all efficient allocations, find the one that gives the coalition receiving the smallest benefit as large a benefit as possible. This is the **maxmin-benefit** allocation, the solution of the LP

$$\varepsilon_1^* \;=\; \max_{x,\,\varepsilon}\; \varepsilon$$

$$\text{subject to} \quad \sum_{i\in N} x_i = C(N), \qquad C(s) - \sum_{i\in N} s_i \cdot x_i \;\geq\; \varepsilon \quad \forall\, s \subsetneq N,\; s \neq \emptyset.$$

The LP finds the largest floor ε such that every coalition receives at least that benefit. It is a well-defined criterion that aims to maximize the stability of cooperation by reducing the worst-case willingness to leave. As a provisional definition, the **nucleolus** is the solution of this LP — the most-egalitarian allocation in the core. This working definition already captures the essential idea and is all that is needed to solve the two-player case; we will see shortly that for larger games it requires a refinement.

Two-player example. The two proper non-empty sub-coalitions are $\{1\}$ and $\{2\}$. Substituting the efficiency constraint $x_2 = 120 - x_1$, the maxmin LP reduces to a one-variable problem, and the two excess functions become affine in x_1 alone:

$$e(\{1\},x) = 90 - x_1, \qquad e(\{2\},x) = 100 - (120 - x_1) = x_1 - 20.$$

The worst-case benefit across the two sub-coalitions is therefore

$$\theta(x_1) \;=\; \min\big(90 - x_1,\;\; x_1 - 20\big),$$

the pointwise minimum of a decreasing line $90 - x_1$ and an increasing line $x_1 - 20$. As depicted in Figure 2, maximizing θ over the core $[20, 90]$ occurs precisely at the *crossover point* of the two lines:

$$90 - x_1^* = x_1^* - 20 \;\implies\; x_1^* = 55, \quad x_2^* = 65, \quad \varepsilon^* = 35.$$

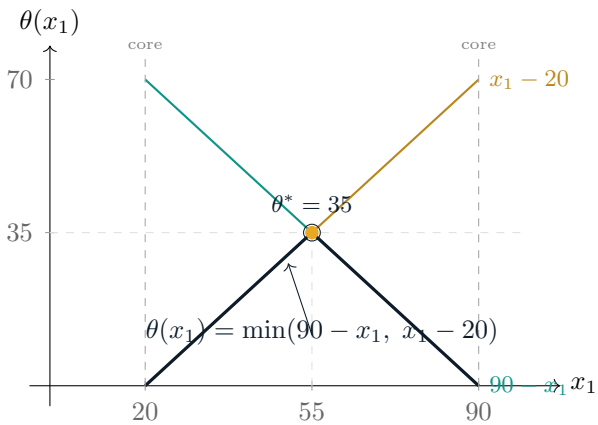

Figure 2: Worst-case benefit $\theta(x_1) = \min(90 - x_1,\ x_1 - 20)$ over the core $x_1 \in [20, 90]$ (navy piecewise-linear curve). The teal line is $e(\{1\}, x) = 90 - x_1$; the gold line is $e(\{2\}, x) = x_1 - 20$. The maximum $\theta^* = 35$ is attained at $x_1^* = 55$, the crossover of the two affine functions — the unique nucleolus of the 2-player game.

Both coalitions receive exactly the same benefit of 35 — an equal split of the total savings of 70.

> **Key insight from the 2-player case.** When only two proper sub-coalitions exist, $\theta(x_1)$ is the minimum of two affine functions: one decreasing (coalition 1 benefits more as x_1 falls) and one increasing (coalition 2 benefits more as x_1 rises). The maxmin allocation is precisely the point where these two lines cross — the *equal-benefit* allocation. For $n = 2$ the maxmin allocation always splits the total savings equally: each coalition saves $(C(\{1\}) + C(\{2\}) - C(N))/2 = 70/2 = 35$.

2.3 Lexicographic Refinement

Room for improvement. In the 2-player case, the maxmin LP had a unique optimizer — the crossover of the two excess lines — so the working definition above pinned down a single allocation. In general this does not happen. With three or more players, the maxmin LP typically has a *set* of optimal allocations: many x can tie at the same worst-case benefit ε_1^*, and the criterion by itself fails to select one. The solution concept has room for improvement, and the refinement should preserve the spirit of maxmin: once the worst-off coalition is as well-off as possible, turn attention to the second-worst, then the third-worst, and so on.

The lexicographic refinement

The nucleolus refines the maxmin idea by applying it *recursively* after each tie is broken:

1. **Raise the floor first.** Among all efficient allocations, find those that maximize the minimum benefit — solve the maxmin LP above. This is the most pressing fairness concern: the coalition receiving the smallest benefit has the strongest incentive to defect.

2. **Among those, raise the second floor.** Once the minimum benefit is pinned, there may still be latitude to increase the second-smallest benefit without reducing the first. Do so.

3. **Continue lexicographically.** At each step, fix the already-optimized excesses and maximize the next floor over the remaining coalitions. The process terminates because each step strictly reduces the dimension of the feasible set.

Formal definition

Let $\theta(x) \in \mathbb{R}^{2^n-2}$ be the vector of excesses of all proper non-empty coalitions, sorted in *increasing* order (worst-off coalition first). The **nucleolus** is the allocation $x^* \in \mathcal{C}$ that *lexicographically maximizes* $\theta(x)$ over the core:

$$x^* \;=\; \operatorname*{lex\,arg\,max}_{x \in \mathcal{C}} \; \theta(x).$$

Sequential-LP computation

The nucleolus is computed via the following sequence of LPs. Step 1 is the maxmin LP already introduced; each subsequent step freezes the currently-worst coalitions and raises the next floor.

Step 1 — maximize the minimum benefit. Solve the maxmin LP above and record ε_1^* and the set $\mathcal{B}_1$ of binding coalitions (those with $e(s,x) = \varepsilon_1^*$ at optimum).

Step k — fix binding coalitions, maximize the next floor. For all $s \in \mathcal{B}_1 \cup \cdots \cup \mathcal{B}_{k-1}$, constrain $e(s,x) \geq \varepsilon_\ell^*$ to the value from the step ℓ at which s became binding. Then maximize ε over the remaining coalitions:

$$\varepsilon_k^* \;=\; \max_{x,\,\varepsilon} \; \varepsilon$$

subject to efficiency, the ensured-excess constraints for the already binding coalitions, and $e(s,x) \geq \varepsilon$ for the remaining s. Repeat until the allocation x^* is uniquely determined.

3 Application: Transmission Design

3.1 Transmission Design Model

Sets and notation. Let $N = \{1, 2, 3, \ldots, n\}$ be the set of n nodes with renewable units and 0 be the high-voltage connecting substation node. We also assume a set of collector nodes N^c; nodes that can be built to aggregate and transmit the power flow of two or more generators to the high-voltage connecting substation, with no local generation injection. The set of candidate lines is L, where each $l \in L$ represents a potential physical connection between any two nodes in $\{0\} \cup N \cup N^c$. For each line l, let $\mathrm{INV}_l > 0$ (in \$/MW.year) be its investment cost per unit of capacity built, and $F_l \geq 0$ (in MW) its built capacity (a decision variable). The tariff $P > 0$ (in \$/MW) is the connection tariff for accessing the high-voltage grid (paid on the total peak power injected into the high-voltage grid per year). The historical data consists of T periods, with renewable generation $g_{i,t} \geq 0$ for unit i and period t (e.g., let's consider hours for the sake of simplicity; in general, it is measured at a higher granularity, such as 15 or 5 minutes).

Coalition indicator. Recall from Section 1 that a coalition $s \subseteq N$ is encoded as a binary vector $s \in \{0,1\}^n$, where $s_i = 1$ means unit i belongs to the coalition s, and zero otherwise. When studying the cost of designing the connection network for a given coalition s, only units within the coalition should be considered to inject generation into the network.

Physical intuition of the objective. The coalition must pay two costs: (i) the infrastructure investment to build enough line capacity to transport the maximum power the coalition will ever generate; and (ii) the high-voltage access tariff, proportional to the maximum simultaneous injection — captured by the variable G, which at optimality equals the peak demand at the substation. Variants that consider self-curtailment decisions, depending on the cost allocated to each generator, can be considered. A related work, focused on forward-contracting risk-sharing — nucleolus-based quota sharing for a renewable energy portfolio via MILP and Benders decomposition — is developed in Freire et al. (2015).

3.2 LP Formulation of the Cost Function

For a given coalition $s \in \{0,1\}^n$ and parameters $(P,\, \mathrm{INV}_l,\, \{g_{i,t}\})$, the coalition cost $C(s)$ is the optimal value of:

$$C(s) \;=\; \min_{\substack{F_l \geq 0,\, l \in L \\ f_{lt} \in \mathbb{R},\, l \in L,\, t \in T \\ G \geq 0}} \quad \sum_{l \in L} \mathrm{INV}_l \cdot F_l \;+\; P \cdot G$$

subject to:

Power flow balance at each generation node $i \in N$, for each period $t \in T$:

$$\sum_{l \in \delta^+(i)} f_{lt} - \sum_{l \in \delta^-(i)} f_{lt} = s_i \cdot g_{i,t} \qquad \forall\, i \in N,\ \forall\, t \in T, \tag{1}$$

where $\delta^+(i)$ is the set of lines leaving node i and $\delta^-(i)$ the set entering it. The right-hand side injects $g_{i,t}$ if unit i is in the coalition ($s_i = 1$), and zero otherwise.

Power flow balance at each intermediate collector node $i \in N^c$, for each period $t \in T$:

$$\sum_{l \in \delta^+(i)} f_{lt} - \sum_{l \in \delta^-(i)} f_{lt} = 0 \qquad \forall\, i \in N^c,\ \forall\, t \in T, \tag{2}$$

where the zero on the right-hand side indicates a transit node.

Power flow balance at the substation 0 ("demand" or extraction node), for each period $t \in T$:

$$\sum_{l \in \delta^-(0)} f_{lt} - \sum_{l \in \delta^+(0)} f_{lt} = \sum_{i \in N} s_i \cdot g_{i,t} \qquad \forall\, t \in T. \tag{3}$$

At this node, the total generation should be set as a demand (extraction) to ensure that power flows from the generator nodes to the high-voltage connecting substation node 0. Note the change in the order of the summation terms compared to the previous expressions.

Line capacity constraints, for each $l \in L$ and $t \in T$:

$$-F_l \leq f_{lt} \leq F_l \qquad \forall\, l \in L,\ \forall\, t \in T. \tag{4}$$

Peak injection lower bound, for each $t \in T$:

$$G \geq \sum_{i \in N} s_i \cdot g_{i,t} \qquad \forall\, t \in T. \tag{5}$$

Non-negativity and free flow:

$$F_l \geq 0 \ \ \forall\, l \in L, \qquad f_{lt} \in \mathbb{R} \ \ \forall\, l,\, t, \qquad G \geq 0. \tag{6}$$

3.3 Structural Insights

Several elegant properties of the model deserve emphasis.

Insight 1 — Decoupling of investment and power flow. F_l is a single number for each line — it represents the *capacity to be built*, not a per-period flow. The constraint $|f_{lt}| \leq F_l$ links the operational problem (flows by period) to the investment decision. At optimality, $F_l^* = \max_t |f_{lt}^*|$, so capacity is driven by the worst-case period.

Insight 2 — G captures peak demand. Since $G \geq \sum_i s_i g_{i,t}$ for all t and $P > 0$, at optimality

$$G^* = \max_{t \in T} \sum_{i \in N} s_i g_{i,t}.$$

The tariff payment is thus $P \cdot \max_t \sum_i s_i g_{i,t}$ — the peak simultaneous renewable output of the coalition.

Insight 3 — LP complexity scales linearly in T. The LP has $|L| + |L| \cdot |T| + 1$ variables and $|N| \cdot |T| + |T| + 2|L| \cdot |T|$ constraints (plus the G bound). It scales *linearly* in T, making it tractable even for large historical datasets — no combinatorial explosion.

3.4 Proof of Core Non-Emptiness via LP Duality and Convexity

We now prove that the core $\mathcal{C}$ is non-empty, using LP duality and the convexity of the coalition cost function $C(s)$. The argument uses LP duality on a coalition-indexed allocation LP to exhibit an element of the core. The key insight is that C is convex as the value function of a parametric LP — a stronger and more natural property than subadditivity, from which Jensen's inequality directly delivers the result.

Step 1 — A relaxed LP whose optimum bounds the core

Define the following LP with $x \in \mathbb{R}^n$ *unrestricted in sign*:

$$\begin{aligned} z \;&=\; \max_{x \in \mathbb{R}^n} \sum_{i \in N} x_i \\ \text{subject to} \quad \sum_{i \in N} s_i \cdot x_i \;&\leq\; C(s) \qquad \forall\, s \subsetneq N,\; s \neq \emptyset. \end{aligned}$$

This LP relaxes the core by dropping the efficiency constraint $\sum_{i \in N} x_i = C(N)$ and instead maximizing the total allocation. We claim $z \geq C(N)$, which immediately implies the core is non-empty: assuming $z \geq C(N)$ (to be proved below), if x^{opt} achieves this value, then the allocation

$$x_i^* \;=\; x_i^{\text{opt}} - \frac{z - C(N)}{n} \qquad \forall\, i \in N$$

redistributes the surplus $\delta = (z - C(N))/n \geq 0$ uniformly, giving $\sum_{i \in N} x_i^* = C(N)$ (efficiency) while preserving the sub-coalition constraints:

$$\sum_{i \in N} s_i \cdot x_i^* = \underbrace{\sum_{i \in N} s_i \cdot x_i^{\text{opt}}}_{\leq C(s)} - \underbrace{|s|}_{\geq 1} \cdot \delta \;\leq\; C(s).$$

Hence $x^* \in \mathcal{C}$. It remains to prove $z \geq C(N)$. Prove it using duality applied to the LP above, together with the main properties of $C(s)$, such as convexity and positive homogeneity.

Comments In the real world, the construction of shared infrastructure aims to exploit economies of scale, materialized as reduced investment costs, complementarity among renewable generation, as their production peaks are not necessarily coincident, and the opportunity cost of energy sold into the market.

The latter introduces an additional dimension, namely the interaction between spot and forward markets. It is not difficult to see that the maximum total flow injected into the high-voltage network for market sale occurs only during very few specific periods. Therefore, the value of this marginal MW sold into the market should be compared with the fixed cost of building the entire infrastructure. This reasoning suggests that it may be optimal not to inject all generated power, as doing so would require infrastructure that is often underutilized. Accordingly, the optimal design of inverters, including well-calibrated clipping of part of the generation, is a well-known practice.

Additionally, another important aspect is the binary (nonconvex) nature of investment decisions. In practice, decisions such as installing a collecting node or a transmission line are discrete rather than continuous, as assumed in our model. This breaks the convexity property. This does not imply that the core is empty, but rather that the problem becomes closer to real-world conditions, for which a general proof may not be available. Nevertheless, the nucleolus remains a powerful tool in this setting. If the nucleolus problem yields a positive floor variable ε^*, this indicates that the core is non-empty and that there is positive synergy from cooperative investment in interconnections.

Another interesting and relevant topic concerns the installation of storage devices. Storage can play multiple roles, ranging from arbitrage in the spot and forward markets to reducing transmission infrastructure costs.

Finally, power flows must satisfy Kirchhoff's Second Law and, depending on voltage levels, account for losses.

This framework can also be extended to consumers, distributed generation, and flexibility markets. As the number of agents increases, the set of constraints grows exponentially. In this case, decomposition approaches similar to those proposed in Freire et al. (2015) become relevant.

References

Lucas Freire, Alexandre Street, Delberis A. Lima, and Luiz A. Barroso. A hybrid MILP and Benders decomposition approach to find the nucleolus quota allocation for a renewable energy portfolio. *IEEE Transactions on Power Systems*, 30(6):3265–3275, November 2015. doi: 10.1109/TPWRS.2014.2374532.